Filter Representation in Vectorized Query Execution (DAMON'21) — What it REALLY is (with examples)

Paper: **"Filter Representation in Vectorized Query Execution"** (Ngom et al., 2021)

If you feel "I don't get the purpose," focus on this sentence:

> A filter representation is a **cheap way to remember which rows are still valid inside a batch**, so operators can keep working without copying data.

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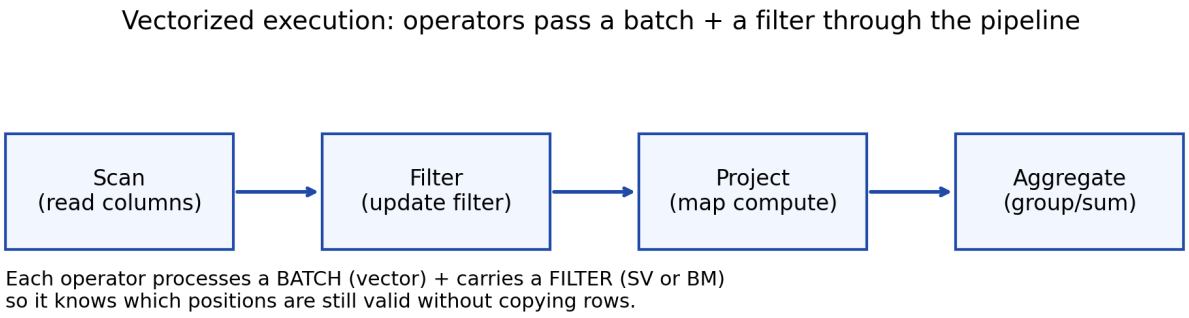
1) The big picture (architecture)

A vectorized DB executes queries as a pipeline of operators. Each operator processes:

- a **batch (vector)** of values (often 1–2k rows)
- plus a **filter** telling it which batch positions are valid

Diagram: batch + filter flows through the pipeline



Key idea:

- The *columns* for the batch (arrays of values) stay in memory.
- The filter changes as the query applies conditions.

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2) Why do we even need this? (the purpose)

When you apply a `WHERE` condition, you have two choices:

Option A: Copy survivors into a new output vector

If a row passes the filter, you physically copy its values into a new vector.

Problem:

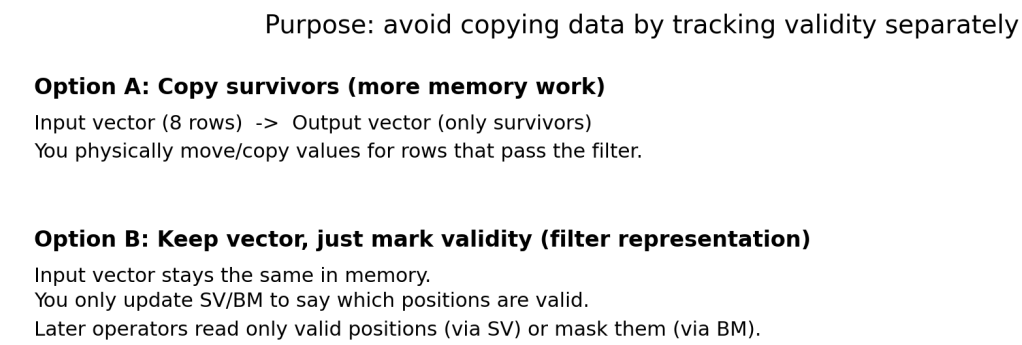
- copying values again and again costs CPU and memory bandwidth

Option B: Don't copy. Keep the vectors and just "mask" invalid rows

You keep the original vectors in place. You only update a small structure that says "these positions are valid."

That structure is the **filter representation** (SV or BM).

Diagram: copy vs mask



This is the whole motivation:

- **copying** is often expensive
- **marking validity** can be cheap

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3) A tiny, concrete example (8-row batch)

Imagine one batch has 8 rows.

Batch positions: `0 1 2 3 4 5 6 7`

Column `amount`:

- `20, 150, 5, 130, 200, 10, 90, 180`

Query step:

``sql WHERE amount > 100 ``

Which positions survive?

- positions **1, 3, 4, 7** survive (150, 130, 200, 180)

Diagram: step-by-step batch filter

Example query step: WHERE amount > 100 (batch of 8)

Batch positions:	`0 1 2 3 4 5 6 7`
amount column:	`20 150 5 130 200 10 90 180`
predicate >100:	`F T F T T F F T`
Selection vector:	**[1, 3, 4, 7]**
Bitmap:	**0 1 0 1 1 0 0 1**

Next operator (e.g., compute tax = amount*1.2) can either:

- Selective: compute only for positions in SV
- Full: compute for all 0..7, then keep results where BM=1

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4) Two ways to represent the same survivors (SV vs BM)

Survivors are: `1, 3, 4, 7`

A) Selection Vector (SV)

A selection vector is a list of surviving positions:

- `SV = [1, 3, 4, 7]`

How operators use it (idea):

- iterate `idx` in `[1,3,4,7]`
- read values like `amount[idx]`

Analogy:

- a **VIP guest list**: you only check the people listed

B) Bitmap (BM)

A bitmap is one bit per position:

- `BM = 0 1 0 1 1 0 0 1`

How operators use it (idea):

- either scan all positions and “mask out” the invalid ones
- or do more complex logic to iterate only the 1s

Analogy:

- **light switches**: 1 = keep, 0 = drop

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5) The second key idea: Selective vs Full compute

Once you have survivors, the next operator might compute something:

``sql SELECT amount * 1.2 AS amount_with_tax ``

There are two strategies:

Strategy 1: Selective compute

Compute only for survivors.

With `SV = [1,3,4,7]` you compute 4 multiplications (not 8).

This is great when **few rows survive**.

Strategy 2: Full compute

Compute for all 8 positions, then keep results only where `BM=1`.

This can be faster when:

- the operation is simple (like `* 1.2`)
- the CPU can do it with SIMD

- scanning 0..7 is a tight simple loop

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6) A more “real” query walkthrough (what happens in a DB)

Take a very normal analytics query:

``sql SELECT country, SUM(amount * 1.2) AS taxed_revenue FROM orders WHERE amount > 100 GROUP BY country; ``

In a vectorized engine, conceptually:

1) **Scan** operator reads columns in batches:

- `amount[]` vector

- `country[]` vector

2) **Filter** operator evaluates a`mount > 100` for the batch:

- it does NOT copy rows

- it updates the filter representation (SV or BM)

3) **Project** operator computes a`mount * 1.2`:

- either selective (only survivors) or full (compute all, then mask)

4) **Aggregate** operator groups and sums only the valid rows:

- uses the filter to skip invalid positions

So the filter representation is like a “do not use these positions” note that travels with the batch.

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7) When is SV better vs BM better? (simple rule)

The paper’s main finding:

- **BM tends to win** when the operator is **SIMD-friendly** (simple arithmetic/comparisons) because full scans + SIMD work well with bitmasks.
- **SV tends to win** for many other operations because iterating survivors is simpler/cheaper than bitmap scanning.

Cheat-sheet

| Situation | Likely better | Why | |---|---|---| | Very few rows survive | SV + selective | Touch only survivors | | Many rows survive + simple numeric work | BM + full | Tight loops + SIMD | | Irregular work (strings, complex UDFs) | SV | Bitmap scanning overhead often doesn’t pay | | Unsure | Benchmark | Data + hardware decide |

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8) Glossary (tiny)

- **Tuple**: row
- **Batch / vector**: a small chunk of rows processed together
- **Filter representation**: data structure telling which batch positions are valid
- **SV (selection vector)**: list of valid positions
- **BM (bitmap)**: 1 bit per position (1 = valid)
- **Selectivity**: fraction of rows that survive
- **SIMD**: CPU doing the same operation on multiple values at once

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9) 20-second recap (purpose)

- Vectorized execution processes data in **batches**.
- After each step, some rows are invalid.
- Instead of copying data, the DB keeps vectors and uses **SV or BM** to mark which positions are valid.
- SV vs BM is a performance tradeoff depending on selectivity + SIMD.